

Home assignment: day 1 → day 4

1. Simplify

$$\sum_{a+b+c=100} \binom{100}{a} \cdot \binom{100-a}{b}.$$

2. 1.9(2) in the old enumeration:

Find explicit formula for the sum

$$1^3 + 2^3 + \dots + n^3.$$

Hint: represent the general term k^3 as the sum of binomial coefficients and then use the hockey stick or Christmas stocking or boomerang!

3. [*] Using hockey stick decomposition find the sums

$$\sum_{k=1}^n 5k^2 + 4k.$$

4. [*] Take a deep breath and calculate

$$\binom{-1}{3}, \quad \binom{-5}{5}, \quad \binom{1.5}{3}, \quad \binom{-1/2}{4}, \quad \binom{1/3}{k}, \quad \binom{-1/2}{n}.$$

5. Write the Taylor expansion at $x = 0$ for

$$\frac{1}{1-x}, \quad (1+x)^{-5}, \quad (1+x)^{-1/2}, \quad (1+3x)^{1/3}, \quad (1-5x)^{1/2}.$$

6. Watch a Veritasium video «Newton changed the game»:

<https://youtu.be/gMlf1ELvRzc?si=VASoVKQj3Pk-wstA>.

You are more than welcome to solve other exercises that we skipped. Starred exercises will be on the quiz next day.